

PSYCH 891: Cognitive and Neural Development

MWF 1:25PM - 2:15PM

Tobin Hall 520

Instructor: Dr. Lauren Aulet

Office Hours: Tuesday 3-4pm, or by appointment (Tobin 425)

Course Description

This course examines the fundamental processes underlying cognitive and neural development from infancy through childhood. Students will explore the theoretical debate between nativism and empiricism, investigating how core cognitive abilities emerge through the interaction of innate mechanisms and environmental experience. The course integrates behavioral and neuroimaging evidence to understand how the developing brain gives rise to sophisticated cognitive abilities.

Assignments

This will be a heavily reading- and discussion-based course. Participation is paramount.

Grade:

20% Participation

25% Individual Presentations (2 presentations / student; 12.5% per presentation)

15% Research Proposal Outline

40% Final Research Proposal

Individual Presentation:

On the first day of class, we will do sign-ups for individual presentations (one student per week; each person will sign up for **TWO** presentations). Each person will pick a paper from the weeks readings (you get to pick one of the 2-3 papers) and present that paper on **MONDAY** of the corresponding week. Presentations are expected to be 15-20 minutes and should cover all key figures in the paper.

Research Proposal:

The research proposal will consist of an introduction, methods, and anticipated results section. The proposal can be on any topic of your choosing (but must be at least somewhat relevant to the course material). The outline will consist of bulleted/outline format version of the final document (e.g., there should be sufficient material to be able to evaluate the rationale and methodology of the project, but need not be in full sentences).

Readings will all be made available on Canvas.

Readings by week:

1. Nativism (Week of Feb 2):

Spelke, E. S., & Kinzler, K. D. (2007). Core knowledge. *Developmental Science*, 10(1), 89-96.

Baillargeon, R. (1987). Object permanence in 3½- and 4½-month-old infants. *Developmental Psychology*, 23(5), 655.

2. Empiricism (Week of Feb 9):

Saffran, J. R., Aslin, R. N., & Newport, E. L. (1996). Statistical learning by 8-month-old infants. *Science*, 274(5294), 1926-1928.

Xu, F., & Kushnir, T. (2013). Infants are rational constructivist learners. *Current Directions in Psychological Science*, 22(1), 28-32.

Smith, L. B., & Thelen, E. (2003). Development as a dynamic system. *Trends in Cognitive Sciences*, 7(8), 343-348.

3. Nativism – Neural (Week of Feb 16):

Reid, V. M., Dunn, K., Young, R. J., Amu, J., Donovan, T., & Reissland, N. (2017). The human fetus preferentially engages with face-like visual stimuli. *Current Biology*, 27(12), 1825-1828.

Kosakowski, H. L., Cohen, M. A., Takahashi, A., Keil, B., Kanwisher, N., & Saxe, R. (2022). Selective responses to faces, scenes, and bodies in the ventral visual pathway of infants. *Current Biology*, 32(2), 265-274.

4. Empiricism – Neural (Week of Feb 23):

Arcaro, M. J., Schade, P. F., Vincent, J. L., Ponce, C. R., & Livingstone, M. S. (2017). Seeing faces is necessary for face-domain formation. *Nature Neuroscience*, 20(10), 1404-1412.

Blumberg, M. S., & Adolph, K. E. (2023). Protracted development of motor cortex constrains rich interpretations of infant cognition. *Trends in Cognitive Sciences*, 27(3), 233-245.

5. Neural specialization (Week of March 2)

Saygin, Z. M., Osher, D. E., Norton, E. S., Youssoufian, D. A., Beach, S. D., Feather, J., ... & Kanwisher, N. (2016). Connectivity precedes function in the development of the visual word form area. *Nature Neuroscience*, 19(9), 1250-1255.

Dehaene, S., & Cohen, L. (2007). Cultural recycling of cortical maps. *Neuron*, 56(2), 384-398.

Kubota, E., Grill-Spector, K., & Nordt, M. (2024). Rethinking cortical recycling in ventral temporal cortex. *Trends in cognitive sciences*, 28(1), 8-17.

6. Neural Lateralization (Week of March 9):

Holowka, S., & Petitto, L. A. (2002). Left hemisphere cerebral specialization for babies while babbling. *Science*, 297(5586), 1515-1515.

Ozernov-Palchik, O., O'Brien, A. M., Lee, E. J., Richardson, H., Romeo, R., Lipkin, B., ... & Fedorenko, E. (2024). Precision fMRI reveals that the language network exhibits adult-like left-hemispheric lateralization by 4 years of age. *bioRxiv*.

7. Neural competition (Week of March 23)

Behrmann, M., & Plaut, D. C. (2015). A vision of graded hemispheric specialization. *Annals of the New York Academy of Sciences*, 1359(1), 30-46.

Dundas, E. M., Plaut, D. C., & Behrmann, M. (2013). The joint development of hemispheric lateralization for words and faces. *Journal of Experimental Psychology: General*, 142(2), 348.

Li, S., Lee, K., Zhao, J., Yang, Z., He, S., & Weng, X. (2013). Neural competition as a developmental process: early hemispheric specialization for word processing delays specialization for face processing. *Neuropsychologia*, 51(5), 950-959.

8. Machine learning (Week of March 30; *Research proposal outlines due Friday April 3rd***):**

Smith, L. B., & Slone, L. K. (2017). A developmental approach to machine learning?. *Frontiers in psychology*, 8, 296143.

Dobs, K., Martinez, J., Kell, A. J., & Kanwisher, N. (2022). Brain-like functional specialization emerges spontaneously in deep neural networks. *Science advances*, 8(11), eabl8913

9. Object recognition (Week of April 6):

Ayzenberg, V., & Behrmann, M. (2024). Development of visual object recognition. *Nature Reviews Psychology*, 3(2), 73-90.

Ayzenberg, V., Sener, S. B., Novick, K., & Lourenco, S. F. (2025). Fast and robust visual object recognition in young children. *Science Advances*, 11(27), eads6821.

10. Spatial Cognition (Week of April 13):

Hermer, L., & Spelke, E. (1996). Modularity and development: The case of spatial reorientation. *Cognition*, 61(3), 195-232.

Frick, A., Möhring, W., & Newcombe, N. S. (2014). Picturing perspectives: development of perspective-taking abilities in 4-to 8-year-olds. *Frontiers in psychology*, 5, 386.

11. Numerical Cognition (Week of April 20):

Izard, V., Sann, C., Spelke, E. S., & Streri, A. (2009). Newborn infants perceive abstract numbers. *Proceedings of the National Academy of Sciences*, 106(25), 10382-10385.

Halberda, J., Mazocco, M. M., & Feigenson, L. (2008). Individual differences in non-verbal number acuity correlate with maths achievement. *Nature*, 455(7213), 665-668.

12. Geometry (Week of April 27):

Dehaene, S., Izard, V., Pica, P., & Spelke, E. (2006). Core knowledge of geometry in an Amazonian indigene group. *Science*, 311, 381-384.

Sablé-Meyer, M., Fagot, J., Caparos, S., van Kerkoerle, T., Amalric, M., & Dehaene, S. (2021). Sensitivity to geometric shape regularity in humans and baboons: A putative signature of human singularity. *Proceedings of the National Academy of Sciences*, 118(16), e2023123118.

Heimler, B., Behor, T., Dehaene, S., Izard, V., & Amedi, A. (2021). Core knowledge of geometry can develop independently of visual experience. *Cognition*, 212, 104716.

13. Symbols (Week of May 4: *Research proposals due Friday May 8th***):**

DeLoache, J. S. (2004). Becoming symbol-minded. *Trends in Cognitive Sciences*, 8(2), 6670.

DeLoache, J. S., Uttal, D. H., & Rosengren, K. S. (2004). Scale errors offer evidence for a perception-action dissociation early in life. *Science*, 304(5673), 1027-1029.

Academic Honesty:

Since UMass Amherst is strongly committed to academic integrity, which is defined as completing all academic work without cheating, lying, stealing, or receiving unauthorized assistance from any other person, or using any source of information not appropriately authorized or attributed. As a community, we hold each other accountable and support each other's knowledge and understanding of academic integrity. Academic dishonesty is prohibited in all programs of the University and includes, but is not limited to: Cheating, fabrication, plagiarism, lying, facilitating dishonesty, via analogue and digital means. Sanctions may be imposed on any student who committed or participated in an academic integrity infraction. Any person who has reason to believe that a student has committed an academic integrity infraction should bring such information to the attention of the appropriate course instructor as soon as possible. All students at the University of Massachusetts Amherst have read and acknowledged the Commitment to Academic Integrity and are knowingly responsible for completing all work with integrity and in

accordance with the policy: <https://www.umass.edu/senate/book/academic-regulations-academic-integrity-policy>

Accommodation:

The University of Massachusetts Amherst is committed to providing an equal educational opportunity for all students. If you have a documented physical, psychological, or learning disability on file with Disability Services (DS), you may be eligible for reasonable academic accommodations to help you succeed in this course. If you have a documented disability that requires an accommodation, please notify me within the first two weeks of the semester so that we may make appropriate arrangements. For further information, please visit Disability Services (<https://www.umass.edu/disability/>)

Diversity:

In recognition and affirmation of the worth and dignity of all persons, the instructor of this class is dedicated to upholding the values of diversity and inclusion and to minimizing disparities in access to learning. All students who are willing to learn are equally welcome without regard to age, race, ethnicity, birthplace, sex, gender identity, disability, sexual orientation or identity, socioeconomic status, political party or beliefs, or other group affiliation or personal characteristics. All students are likewise expected to accord each other the same level of respect, bearing in mind that becoming more respectful and inclusive is a journey rather than a destination. We will all have stumbling points along the way, including the instructor, and these are often valuable learning opportunities. If I ever say or do anything insensitive in this class, I sincerely welcome students to call me out or communicate their discomfort to me in person after class, during office hours, by email, or anonymously.

Title IX:

In accordance with Title IX of the Education Amendments of 1972 that prohibits gender-based discrimination in educational settings that receive federal funds, the University of Massachusetts Amherst is committed to providing a safe learning environment for all students, free from all forms of discrimination, including sexual assault, sexual harassment, domestic violence, dating violence, stalking, and retaliation. This includes interactions in person or online through digital platforms and social media. Title IX also protects against discrimination on the basis of pregnancy, childbirth, false pregnancy, miscarriage, abortion, or related conditions, including recovery. There are resources here on campus to support you. A summary of the available Title IX resources (confidential and non-confidential) can be found at the following link: <https://www.umass.edu/titleix/resources>. You do not need to make a formal report to access them. If you need immediate support, you are not alone. Free and confidential support is available 24 hours a day / 7 days a week / 365 days a year at the SASA Hotline 413-545-0800.